

Vitamin K1 (phytonadione, phytomenadione): Drug information - UpToDate

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For additional information see "Vitamin K1 (phytonadione, phytomenadione): Patient drug information" and "Vitamin K1 (phytonadione, phytomenadione): Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

ALERT: US Boxed Warning

Hypersensitivity reactions with IV and IM use:

Fatal hypersensitivity reactions, including anaphylaxis, have occurred during and immediately after IV and IM injection of phytonadione. Reactions have occurred despite dilution to avoid rapid IV infusion and upon first dose. Avoid the IV and IM routes of administration unless the SUBQ route is not feasible and the serious risk is justified.

Brand Names: US

- Mephyton

Brand Names: Canada

- AquaMEPHYTON;
- Konakion;
- Mephyton

Pharmacologic Category

- Vitamin, Fat Soluble

Dosing: Adult

Dosage guidance:

Safety: Due to risk of a severe infusion reaction, including anaphylaxis, the maximum rate of IV administration is 1 mg/minute.

Reversal of anticoagulation due to warfarin or other coumarin or indandione derivatives

Reversal of anticoagulation due to warfarin or other coumarin or indandione derivatives:

Note: Dose and route depend on the severity of bleeding and/or INR.

Urgent reversal: IV administration is recommended. For life-threatening bleeding, vitamin K should be administered in combination with clotting factors (eg, 4-factor prothrombin complex concentrate [PCC], or fresh frozen plasma if 4-factor PCC is not available) (Ref). Following IV administration, reversal begins in ~2 hours with expected full reversal within 24 hours (Ref).

Nonurgent reversal: Oral administration is recommended to reduce INR in ~24 to 48 hours (Ref).

Avoid other routes: SUBQ administration is not recommended due to unpredictable absorption (Ref). Avoid IM administration due to risk of hematoma formation.

Usual dosage range:

Oral: Initial: 2.5 to 10 mg, depending on the INR. Administer as a single dose. Measure INR after 12 to 48 hours and administer another dose as needed.

IV: Initial: 2.5 to 10 mg, depending on the INR and severity of bleeding. Administer as a single dose over 10 to 20 minutes (maximum infusion rate: 1 mg/minute). Measure INR after 6 to 12 hours and administer another dose as needed.

Note: High doses of vitamin K (eg, >10 to 15 mg) may cause warfarin resistance for ≥ 1 week; if anticoagulation is needed, an alternative agent may be necessary (Ref).

Reversal of Anticoagulation From Warfarin

Indication

Management considerations

Vitamin K administration

^a 1 mg oral dose may be administered using 0.5 mL of 2 mg/mL parenteral preparation by mouth.

^b Patriquin 2011.

^c ACCP [Hirsh 2008].

^d ACCP [Holbrook 2012].

^e ASH [Witt 2018].

^f AHA/ACC [Nishimura 2014]; ACC/AHA [Otto 2021].

^g Farrow 2020.

^h PCC=prothrombin complex concentrate.

ⁱ ACCP [Ageno 2012].

^j NCS/SCCM [Frontera 2016].

^k ACCP [Douketis 2022].

^l Douketis 2021.

^m Garcia 2022.

ⁿ ACC [Doherty 2017].

^o Nkomo 2023.

Supratherapeutic INR and no evidence of bleeding

INR above therapeutic range but <4.5

Consider holding the next dose of warfarin and/or reduce maintenance dose; increase INR monitoring.

Elevated INR may be due to transient factors, so warfarin dose reduction is not always necessary.

Routine administration of vitamin K is not recommended.^b

INR 4.5 to 10

Discontinue warfarin, monitor INR frequently, and resume an appropriately reduced warfarin dose when INR is in desired range.

May consider administering oral vitamin K if additional risk factors for bleeding exist.

Routine administration of vitamin K is not recommended.

If administered, one dose of **oral** vitamin K 1 to 2.5 mg is recommended.^{a,c,d,e,f}

INR >10

Discontinue warfarin, monitor INR frequently, and resume an appropriately reduced warfarin dose when INR is in desired range.

Consider administering oral vitamin K depending on bleeding and thrombotic risks.

If administered, one dose of **oral** vitamin K 2.5 to 5 mg is usually recommended; recheck INR after 12 to 24 hours; may administer a second dose if necessary.^{c,d}

For patients with a mechanical prosthetic heart valve, a lower **oral** vitamin K dose of 1 to 2.5 mg may be administered to avoid overcorrection of INR.^{a,f}

Some data suggest that vitamin K is not routinely needed in the absence of bleeding.^g

Bleeding

Minor bleeding

Discontinue warfarin, monitor INR frequently, and resume warfarin at an appropriately adjusted dose when it is safe to do so.

Consider administering oral vitamin K depending on INR, site of bleeding, risk of progression to more serious bleeding, and thrombotic risk.

If administered, **oral** vitamin K 2.5 to 5 mg is recommended; if INR remains elevated after 24 hours, may administer another dose.^b

Major bleeding into a critical site and/or life-threatening bleeding (including intracranial hemorrhage)

Discontinue warfarin and urgently administer a 4-factor PCC^h in combination with IV vitamin K; monitor INR frequently and assess for hemostasis.

Administer **IV** vitamin K 10 mg over 10 to 20 minutes (maximum infusion rate: 1 mg/minute) as soon as possible in combination with a 4-factor PCC; if INR remains elevated after 12 to 24 hours, may administer another dose of vitamin K.^{c,d,i,j}

Invasive procedure or surgery

Elective

Thrombotic and bleeding risks vary depending on individual circumstances. Clinical practice guidelines may help guide whether to interrupt warfarin therapy, how long before an elective procedure or surgery warfarin should be discontinued (if necessary), whether bridging with a parenteral anticoagulant is needed, and when to restart warfarin therapy after the procedure/surgery.^{k,n}

If it is decided to discontinue warfarin, therapy is typically stopped 5 days before the procedure/surgery. If INR remains >1.5 the day before the procedure/surgery, oral vitamin K may be considered but is not recommended routinely. Recheck INR on the day of the procedure/surgery.^{l,k}

Administer **oral** vitamin K 1 to 2.5 mg once if needed the day before the procedure/surgery.^{a,b,l}

Urgent

For urgent procedures or surgeries in patients at high bleeding risk, vitamin K with or without a 4-factor PCC may be needed.

Administer **IV** vitamin K 1 to 10 mg over 10 to 20 minutes (maximum infusion rate: 1 mg/minute) with or without a 4-factor PCC, depending on urgency of the procedure/surgery and risk of bleeding.^{l,m}

For patients with a mechanical prosthetic heart valve, some experts recommend low-dose **oral or IV** vitamin K (ie, 1 or 2 mg) in combination with a 4-factor PCC in order to balance risk of bleeding and valve thrombosis.⁹

Refer to institutional policies and procedures.

Reversal of anticoagulation due to long-acting anticoagulant rodenticides

Reversal of anticoagulation due to long-acting anticoagulant rodenticides: Consultation with a clinical toxicologist or poison control center and hematologist is highly recommended when making treatment decisions, including dose, route, and duration of therapy (Ref).

No significant bleeding or minor bleeding (eg, epistaxis, ecchymosis):

Note: Ideal dosing has not been established.

INR <4.5 and no bleeding: No vitamin K required; monitor and observe closely (Ref).

INR ≥4.5 or minor bleeding: **Oral:** Some experts recommend an initial dose of 10 mg per day in otherwise healthy patients; titrate as needed based on serial PT and INR measurements (eg, initially every 48 to 72 hours) (Ref). The reported daily dosage range is variable (up to 800 mg per day (Ref)). Patients may require treatment for several months and should be under the care of a hematologist (Ref); compliance with treatment and monitoring is of utmost importance during outpatient therapy (Ref).

Serious bleeding (eg, intracranial, GU, GI bleeding):

Note: Ideal dosing has not been established. For life-threatening bleeding, vitamin K should be administered in combination with clotting factors (eg, 4-factor prothrombin complex concentrate [PCC] or 3-factor PCC in combination with either recombinant factor VIIa or fresh frozen plasma if 4-factor PCC is not available) (Ref). Treatment should begin with IV vitamin K and then transition to oral therapy (Ref).

IV, Oral: Reported initial doses are variable and range from 10 to 300 mg per day; titrate as needed based on serial PT and INR measurements (Ref). Clinicians may choose to initiate therapy according to recommendations for reversal of anticoagulation due to warfarin or other coumarin or indandione derivatives at doses indicated for major bleeding at any INR elevation (Ref). The reported daily dosage range is variable (up to 800 mg per day (Ref)). Patients may require treatment with oral vitamin K for several months and should be under the care of a hematologist (Ref); compliance with treatment and monitoring is of utmost importance during outpatient therapy (Ref).

Vitamin K deficiency

Vitamin K deficiency (without liver disease):

Note: Vitamin K deficiency in healthy adults is rare; deficiency tends to be associated with specific conditions (eg, bariatric surgery, biliary disease, malnutrition, cystic fibrosis, prolonged use of broad-spectrum antibiotics) (Ref). Avoid oral route in situations associated with GI malabsorption.

Treatment of coagulopathy:

IV, Oral, SUBQ: Usual dose: 10 mg once; may repeat in 48 to 72 hours if coagulopathy persists (Ref); range: 1 to 25 mg (manufacturer's labeling); maximum IV infusion rate: 1 mg/minute (Ref).

Note: SUBQ administration is not recommended in patients with generalized edema (eg, anasarca) (Ref).

Bariatric surgery:

Prevention of vitamin K deficiency, supplementation:

Laparoscopic adjustable gastric band, Roux-en-Y gastric bypass, or sleeve gastrectomy: **Oral:** 90 to 120 mcg once daily (Ref).

Biliopancreatic diversion/duodenal switch: **Oral:** 300 mcg once daily (Ref).

Acute vitamin K repletion for malabsorption:

IV: 10 mg once over 10 to 20 minutes (Ref).

Chronic vitamin K repletion for malabsorption:

IV: 1 to 2 mg once weekly (Ref).

Oral: 1 to 2 mg once daily (Ref).

Note: In the United States and Canada, low-dose oral preparations can be found over the counter. A pharmacist can help the patient locate a low-dose supplement from a reliable supplier or prepare a compounded oral solution.

Cystic fibrosis with pancreatic insufficiency:

Prevention of vitamin K deficiency, supplementation:

Oral: 2.5 to 5 mg once weekly. **Note:** Additional supplementation may be required during antibiotic therapy (Ref).

Liver disease, treatment of coagulopathy

Liver disease, treatment of coagulopathy (off-label use):

Note: Consider use if vitamin K deficiency is suspected (eg, due to poor nutrition, cholestatic disease, diarrheal illness, or antibiotic use); vitamin K does not correct clotting factor deficiency due to parenchymal liver dysfunction and is not typically used for variceal bleeding (Ref).

Major bleeding (nonvariceal): IV: 10 mg once over 10 to 20 minutes (maximum infusion rate: 1 mg/minute) (Ref).

Minor bleeding (nonvariceal):

Oral: 10 mg once daily for 3 days (Ref).

IV: 10 mg once over 10 to 20 minutes (maximum infusion rate: 1 mg/minute) (Ref).

Nonbleeding: Oral, IV, SUBQ: 5 to 10 mg once (maximum IV infusion rate: 1 mg/minute). **Note:** Oral route is preferred unless there is concern for poor absorption (Ref).

Warfarin-associated INR variation due to vitamin K deficiency

Warfarin-associated INR variation due to vitamin K deficiency (off-label use):

Note: For use in patients with poor INR control due to suspected vitamin K deficiency (eg, insufficient dietary intake). Ensure proper warfarin adherence and consult with the anticoagulation specialist managing the patient before prescribing.

Oral: 100 to 200 mcg once daily (Ref).

Note: In the United States and Canada, low-dose (eg, 100 mcg) oral preparations can be found over the counter. A pharmacist can help the patient locate a low-dose supplement from a reliable supplier or prepare a compounded oral solution.

Dosing: Kidney Impairment: Adult

The renal dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

Altered kidney function: No dosage adjustment necessary for any degree of kidney impairment (no unmetabolized vitamin K eliminated in urine) (Ref).

Hemodialysis, intermittent (thrice weekly): Unlikely to be significantly dialyzed (lipophilic characteristics) (Ref): No supplemental dose or dosage adjustment necessary (Ref).

Peritoneal dialysis: Unlikely to be significantly dialyzed (lipophilic characteristics): No dosage adjustment necessary (Ref).

CRRT: No dosage adjustment necessary (Ref).

PIRRT (eg, sustained, low-efficiency diafiltration): No dosage adjustment necessary (Ref).

Dosing: Liver Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling.

Dosing: Older Adult

Refer to adult dosing.

Dosing: Pediatric

(For additional information see "Vitamin K1 (phytonadione, phytomenadione): Pediatric drug information")

Dosage guidance:

Safety: Dosing presented in **mcg** and mg; verify dosing units.

Clinical considerations: Route of administration varies with indication; careful evaluation of route of administration is important. The IM route should be avoided due to the risk of hematoma formation except in the prevention/treatment of vitamin K deficiency bleeding (Ref).

Parenteral nutrition, maintenance requirement

Parenteral nutrition, maintenance requirement: Limited data available (Ref):

Note: Patients receiving warfarin may not require TPN supplementation of phytonadione.

Infants: IV: 10 **mcg**/kg/day as an additive to parenteral nutrition solution.

Children and Adolescents: IV: 200 **mcg**/day as an additive to parenteral nutrition solution.

Reversal of vitamin K antagonists

Reversal of vitamin K antagonists (eg, warfarin): Limited data available:

Weight-based dosing (preferred): *Chest* recommendations: Infants, Children, and Adolescents: IV: 0.03 mg/kg/dose is recommended for excessively prolonged INR (usually INR >8; no evidence of bleeding) due to vitamin K-antagonist (eg, warfarin): if significant bleeding, consider use of fresh frozen plasma, prothrombin complex concentrates, or recombinant factor VIIa (Ref).

Fixed dosing: **Note:** Smaller pediatric patients should receive doses on the low end of dosing range; excessive dosages may cause warfarin-resistance (Ref).

Infants, Children, and Adolescents:

No bleeding, rapid reversal needed, patient will require further oral anticoagulant therapy: SUBQ, IV: 0.5 to 2 mg.

No bleeding, rapid reversal needed, patient will not require further oral anticoagulant therapy: SUBQ, IV: 2 to 5 mg.

Significant bleeding, not life-threatening: SUBQ, IV: 0.5 to 2 mg; **Note:** Use in combination with fresh frozen plasma.

Significant bleeding, life-threatening: SUBQ, IV: 5 mg; **Note:** Consider use with prothrombin complex concentrate containing factors II, VII, IX, X.

Vitamin K deficiency, prevention, and supplementation

Vitamin K deficiency, prevention, and supplementation (disease-specific): Limited data available:

Biliary atresia (Ref): **Note:** Dose and route are determined by INR value:

Infants 1 to 6 months:

INR >1.2 to 1.5: 2.5 mg once daily orally.

INR >1.5 to 1.8: Initial: 2 to 5 mg IM once followed by 2.5 mg once daily orally.

INR >1.8: Initial: 2 to 5 mg IM once followed by 5 mg once daily orally.

Cholestasis: Infants, Children, and Adolescents: Oral: 2.4 to 15 mg/day (Ref).

Cystic fibrosis: Infants, Children, and Adolescents: Oral: 0.3 to 0.5 mg/day (Ref).

Liver disease: Infants, Children, and Adolescents: Oral: 2.5 to 5 mg/day (Ref).

Dosing: Kidney Impairment: Pediatric

There are no dosage adjustments provided in manufacturer's labeling.

Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in manufacturer's labeling; dosing adjustment unlikely necessary.

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified.

Frequency not defined:

Cardiovascular: Chest pain, flushing, hypotension, tachycardia, weak pulse

Central nervous system: Dizziness

Dermatologic: Diaphoresis, eczematous rash, erythema, erythematous rash, pruritic plaques of the skin, urticaria

Gastrointestinal: Dysgeusia

Hepatic: Hyperbilirubinemia

Hypersensitivity: Anaphylactoid reaction, anaphylaxis, hypersensitivity reaction

Local: Injection site reaction (including pain, swelling, tenderness)

Respiratory: Cyanosis, dyspnea

Miscellaneous: Lesion (scleroderma-like)

Contraindications

Hypersensitivity to phytonadione or any component of the formulation

Warnings/Precautions

Concerns related to adverse effects:

- Dermatologic toxicity: Cutaneous reactions have occurred after parenteral administration, including delayed-type hypersensitivity reactions, eczematous reactions, scleroderma-like patches, and urticaria; onset may occur within 1 day to a year. If skin reactions occur, discontinue therapy and treat appropriately.
- Hypersensitivity/anaphylactoid reactions: Hypersensitivity reactions, including anaphylaxis, chest pain, cyanosis, diaphoresis, dyspnea, flushing, cardiorespiratory arrest, shock, tachycardia, weakness, and death, have occurred. Anaphylaxis typically, but not always, occurred when patients received large IV doses administered rapidly with formulations containing polyethoxylated castor oil (also called polyoxyethylated castor oil); proper dosing, dilution, and administration will minimize risk (ACCP [Agno 2012]; Britt 2018; Riegert-Johnson 2002).

Disease-related concerns:

- Anticoagulant-induced coagulopathy: In patients receiving a therapeutic vitamin K antagonist (VKA) (eg, warfarin), administer a dose of phytonadione that will quickly lower the INR into a safe range without causing resistance to warfarin. High phytonadione doses (eg, >10 to 15 mg) may lead to warfarin resistance for ≥ 1 week.

- Liver disease, coagulopathy: If initial doses do not reverse coagulopathy, then higher doses are unlikely to have any effect. **Note:** Ineffective in hereditary hypoprothrombinemia.
- Long-acting anticoagulant rodenticide (LAAR) ingestion: Patients with LAAR-induced coagulopathy require much larger doses and longer treatment durations (up to months) after exposure compared to that needed to reverse warfarin-induced coagulopathy (Devgun 2020; Gunja 2011).

Special populations:

- Neonates: Use with caution in neonates, especially premature infants; severe hemolytic anemia, jaundice, and hyperbilirubinemia have been reported with larger than recommended doses (10 to 20 mg).

Dosage form specific issues:

- Aluminum: The parenteral product may contain aluminum; toxic aluminum concentrations may be seen with high doses, prolonged use, or renal dysfunction. Premature neonates are at higher risk due to immature renal function and aluminum intake from other parenteral sources. Parenteral aluminum exposure of >4 to 5 mcg/kg/day is associated with CNS and bone toxicity; tissue loading may occur at lower doses (Federal Register 2002). See manufacturer's labeling.
- Benzyl alcohol and derivatives: Some dosage forms may contain benzyl alcohol; large amounts of benzyl alcohol (≥ 99 mg/kg/day) have been associated with a potentially fatal toxicity ("gasping syndrome") in neonates; the "gasping syndrome" consists of metabolic acidosis, respiratory distress, gasping respirations, CNS dysfunction (including convulsions, intracranial hemorrhage), hypotension, and cardiovascular collapse (AAP ["Inactive" 1997]; CDC 1982); some data suggests that benzoate displaces bilirubin from protein binding sites (Ahlfors 2001); avoid or use dosage forms containing benzyl alcohol with caution in neonates. See manufacturer's labeling.
- Polyoxyethylated castor oil: Some injectable dosage forms contain polyoxyethylated castor oil (Cremophor EL) which is associated with hypersensitivity reactions.
- Polysorbate 80: Some dosage forms may contain polysorbate 80 (also known as Tweens). Hypersensitivity reactions, usually a delayed reaction, have been reported following exposure to pharmaceutical products containing polysorbate 80 in certain individuals (Isaksson 2002; Lucente 2000; Shelley 1995). Thrombocytopenia, ascites, pulmonary deterioration, and renal and hepatic failure have been reported in premature neonates after receiving parenteral products containing polysorbate 80 (Alade 1986; CDC 1984). See manufacturer's labeling.

Other warnings/precautions:

- Appropriate route: Oral administration is the safest and requires the presence of bile salts for absorption. In obstructive jaundice or with biliary fistulas, concurrent administration of bile salts would be necessary for proper absorption. Manufacturers recommend the SubQ route over other parenteral routes, however, SubQ is less predictable when compared to the oral route, and efficacy may be delayed. Therefore, the SubQ route has fallen out of favor due to erratic and unpredictable absorption (Crowther 2002; DeZee 2006; Raj 1999). The IV route may be used in selected **nonbleeding** patients and should be used in patients with **major bleeding** due to warfarin-associated coagulopathy (ACCP [Guyatt 2012]; Patriquin 2011). IM route should be avoided due to the risk of hematoma formation (except, for example, in the neonatal and pediatric populations for prevention/treatment of vitamin K deficiency bleeding). Efficacy (eg, control of bleeding, decrease in INR) is delayed regardless of route of administration; patient management may require other treatments in the interim.

Dosage Forms Considerations

Injectable products may contain alcohol, benzyl alcohol, polysorbate 80, propylene glycol, or polyoxyethylated/polyethoxylated castor oil (Cremophor EL).

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Injection, aqueous colloidal: 1 mg/0.5 mL (0.5 mL); 10 mg/mL (1 mL)

Injection, aqueous colloidal [preservative free]: 1 mg/0.5 mL (0.5 mL)

Tablet, oral:

Mephyton: 5 mg [scored]

Generic: 100 mcg, 5 mg

Generic Equivalent Available: US

Yes

Pricing: US

Capsules (K1-1000 Oral)

1 mg (per each): \$0.11

Solution (Phytonadione Injection)

1 mg/0.5 mL (per 0.5 mL): \$4.51 - \$29.68

10 mg/mL (per mL): \$26.40 - \$51.32

Tablets (Phytonadione Oral)

5 mg (per each): \$8.59 - \$66.99

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Extemporaneous Preparations

1 mg/mL Oral Suspension

A 1 mg/mL oral suspension may be made with tablets. Crush six 5 mg tablets in a mortar and reduce to a fine powder. Add 5 mL each of water and methylcellulose 1% and mix to a uniform paste. Mix while adding sorbitol in incremental proportions to **almost** 30 mL; transfer to a calibrated bottle, rinse mortar with sorbitol, and add quantity of sorbitol sufficient to make 30 mL. Label "shake well" and "refrigerate". Stable for 3 days (Ref).

Note: The parenteral formulation may also be used for small oral doses (eg, 1 mg) or situations in which tablets cannot be swallowed (Ref); may administer undiluted or diluted in a beverage (eg, orange juice) (Ref). Phytonadione IV solution administered orally has been shown to be as effective as orally administered tablets (Ref).

Administration: Adult

IV: After dilution of dose in a minimum of 50 mL of compatible solution, administer slowly using an infusion pump over 10 to 20 minutes depending on dose at a rate not to exceed 1 mg/minute (Ref). The injectable route should be used only if the oral route is not feasible or there is a greater urgency to reverse anticoagulation. The IV route may be used in selected nonbleeding patients and should be used in patients with major bleeding due to warfarin-associated coagulopathy (Ref).

SUBQ: SUBQ route is generally not recommended due to erratic and unpredictable absorption (Ref).

IM: IM route should be avoided due to the risk of hematoma formation.

Oral: The parenteral formulation may also be used for small oral doses (eg, 1 mg) or situations in which tablets cannot be swallowed (Ref); may administer undiluted or diluted in a beverage (eg, orange juice) (Ref).

Preparation for Administration: Adult

Dilute injection solution in preservative-free NS, D5W, or D5NS; protect from light. To reduce the incidence of anaphylaxis upon IV administration, dilute dose in a minimum of 50 mL of a compatible solution (Ref).

Administration: Pediatric

Oral: May be administered with or without food. The parenteral formulation may also be used for small oral doses (eg, 1 mg) or situations in which tablets cannot be swallowed (Ref).

Parenteral: **Note:** Limit IV administration to situations where an alternative route of administration is not feasible and the benefit of therapy outweighs the risk of hypersensitivity reactions; proper dosing, dilution, and administration will minimize risk (Ref). Allergic reactions have also occurred with IM and SUBQ injections, albeit less frequently.

SUBQ: Administer undiluted.

IM: Administer undiluted; for use in neonatal patients, verify appropriate concentration (1 mg/0.5 mL).

IV: After dilution, infuse slowly. In pediatric patients, IV doses have been infused over 10 to 30 minutes (Ref); maximum rate of infusion: 1 mg/minute (Ref).

Preparation for Administration: Pediatric

Parenteral: IV: Dilute phytonadione injection in preservative-free NS, D5W, or D5NS. To reduce the incidence of anaphylactoid reaction upon IV administration dilution of dose is recommended (eg, in adults, dilution of the dose in a minimum of 50 mL of compatible solution has been used) (Ref).

Storage/Stability

Parenteral: Store intact ampules and vials in original container at 20°C to 25°C (68°F to 77°F); protect from light.

Tablets: Store in original container at 20°C to 25°C (68°F to 77°F); excursions permitted to 15°C to 30°C (59°F to 86°F). Protect from light.

Pharmacy supply of emergency antidotes: Guidelines suggest that at least 50 to 100 mg of phytonadione be stocked, including both oral and parenteral formulations. Suggested amount is stated to be a sufficient quantity to treat 1 patient weighing 100 kg for an initial 8- to 24-hour period (Dart 2018); actual amount to be stocked should take into account site-specific and population-specific needs.

Use: Labeled Indications

Reversal of anticoagulation due to warfarin or other coumarin or indandione derivatives (eg, brodifacoum): Treatment of anticoagulant-induced deficiency of vitamin K-dependent clotting factors caused by coumarin or indandione derivatives, including warfarin and long-acting anticoagulant rodenticides (eg, brodifacoum, bromadiolone, difenacoum) (Card 2014; Devgun 2020; Feinstein 2020; Gunja 2011).

Vitamin K deficiency (without liver disease): Treatment of deficiency secondary to factors limiting absorption or synthesis of vitamin K (eg, obstructive jaundice, biliary fistula, sprue, ulcerative colitis, celiac disease, intestinal resection, cystic fibrosis of the pancreas, regional enteritis) or drug-induced interference with vitamin K metabolism (eg, salicylates, antibacterial therapy).

Vitamin K deficiency bleeding (formerly known as hemorrhagic disease) of the newborn: Prophylaxis and treatment of vitamin K deficiency bleeding (formerly known as hemorrhagic disease) of the newborn (injection only).

Use: Off-Label: Adult

Liver disease, treatment of coagulopathy; Warfarin-associated INR variation due to vitamin K deficiency

Medication Safety Issues

Sound-alike/look-alike issues:

Mephyton may be confused with melphalan, methadone

Metabolism/Transport Effects

None known.

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within “CYP3A4 Inducers [Strong]” are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the “Launch drug interactions program” link above.

Orlistat: May decrease absorption of Vitamins (Fat Soluble). Management: Administer oral fat soluble vitamins at least 2 hours before or 2 hours after the administration of orlistat. Avoid concomitant administration due to the risk of impaired vitamin absorption. *Risk D: Consider Therapy Modification*

Vitamin K Antagonists: Phytonadione may decrease anticoagulant effects of Vitamin K Antagonists. *Risk C: Monitor*

Pregnancy Considerations

Phytonadione crosses the placenta in limited concentrations (Kazzi 1990).

The dietary requirements of vitamin K are the same in pregnant and nonpregnant women (IOM 2001). In general, medications used as antidotes should take into consideration the health and prognosis of the mother; antidotes should be administered to pregnant women if there is a clear indication for use and should not be withheld because of fears of teratogenicity (Bailey 2003). Use of preservative free solutions are preferred when the injection is needed during pregnancy.

Breastfeeding Considerations

Phytonadione is present in breast milk.

Small amounts of dietary vitamin K can be detected in breast milk and the dietary requirements of vitamin K are the same in breastfeeding and non-breastfeeding women (IOM 2001). According to the manufacturer, the decision to continue or discontinue breastfeeding during therapy should take into account the risk of infant exposure, the benefits of breastfeeding to the infant, and benefits of treatment to the mother. Use of preservative free solutions are preferred when the injection is needed in breastfeeding women.

Dietary Considerations

Dietary adequate intake (IOM 2001; Vanek 2012):

1 to 6 months: 2 **mcg/day**

7 to 12 months: 2.5 **mcg/day**

1 to 3 years: 30 **mcg/day**

4 to 8 years: 55 **mcg/day**

9 to 13 years: 60 **mcg/day**

14 to 18 years: 75 **mcg/day**

>18 years: Males: 120 **mcg/day**; Females: 90 **mcg/day**

Monitoring Parameters

Long-acting anticoagulant rodenticide poisoning: PT/INR; consider serial measurement of plasma coumarin concentration (eg, plasma brodifacoum concentrations) to help guide treatment duration decisions. At least 3 samples collected over the period of 1 week has been suggested to determine treatment duration; an additional measurement taken 2 weeks after treatment discontinuation is recommended to rule out rebound elevation and ensure maintenance of safe concentrations (Nosal 2021).

Mechanism of Action

Promotes liver synthesis of clotting factors (II, VII, IX, X); however, the exact mechanism as to this stimulation is unknown. Menadiol is a water soluble form of vitamin K; phytonadione has a more rapid and prolonged effect than menadione; menadiol sodium diphosphate (K_4) is half as potent as menadione (K_3).

Pharmacokinetics (Adult Data Unless Noted)

Onset of action: Increased coagulation factors: Oral: 6 to 10 hours; IV: 1 to 2 hours.

Peak effect: INR values return to normal: Oral: 24 to 48 hours; IV: 12 to 14 hours.

Absorption: Oral: From intestines in presence of bile; SUBQ: Variable; IM: Readily.

Metabolism: Rapidly hepatic.

Excretion: As metabolites in urine and feces.

Patient Counseling Points

(For additional information see "Vitamin K1 (phytonadione, phytomenadione): Patient drug information")

What is this drug used for?

- It is used to replace low vitamin K.
- It is used to undo the effects of certain blood thinners like warfarin.
- It is used to treat or prevent bleeding.
- It is often given to newborn babies.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

Injection:

- Pain, redness, or swelling where the shot was given
- Change in taste

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

All products:

- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Injection:

- Flushing
- Feeling tired or weak

- Chest pain or pressure, a fast heartbeat, or an abnormal heartbeat
- Dizziness or passing out
- Shortness of breath
- Sweating a lot
- Blue or gray color of the skin, lips, nail beds, fingers, or toes

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Konakion mm;
- (AR) Argentina: Konakion mm | Vitamina k surar pharma | Vitamina k1 biol;
- (AT) Austria: Konakion;
- (BD) Bangladesh: Konakion;
- (BE) Belgium: Konakion;
- (BG) Bulgaria: Konakion;
- (BR) Brazil: Kanakion | Kanakion MM | Kavit | Vikatron | Vita K | Vitamina k;
- (CH) Switzerland: Konakion;
- (CL) Chile: Fitoquinona | Konakion;
- (CN) China: Kai nai jin | Vitamin K1;
- (CO) Colombia: Fitomenadiona | Vitamina k | Vitamina K1;
- (CZ) Czech Republic: Kanavit;
- (DE) Germany: Kanavit | Vitamin K1;
- (DO) Dominican Republic: Konakion;
- (EC) Ecuador: Fitomenadiona | Konakion | Vitamina k;
- (EE) Estonia: Menadion medic;
- (EG) Egypt: Adcockion | Epikavit | Haemokion | K Dion | K sabaa | Konakion | Origin k1 | Phytomenadion | Vitadione;
- (GB) United Kingdom: Konakion mm;
- (GR) Greece: Konakion;
- (HK) Hong Kong: Vitamin K1;
- (ID) Indonesia: Kaywan | Neo k;
- (IE) Ireland: Konakion mm;
- (IN) India: Kenadion;
- (IT) Italy: Konakion;
- (JP) Japan: Dekaywan | Ersupa k1 | Hymeron k1 | Kaywan | Keipole | Mephyton | Nichivita k1 | Onekey | Phytomenadione merck | Phytomenadione towa | Univitan k1 universal | Univitan k1 yoshindo | Vita k1 kobayashi | Vitamin k1 amel | Vitamin k1 daiko | Vitamin k1 fuso | Vitamin k1 meiji |

Vitamin k1 mita | Vitamin k1 nipro | Vitamin k1 nisshin kyorin sei | Vitamin k1 pfizer | Vitamin k1 sawai | Vitamin k1 seiko | Vitamin k1 showa | Vitamin k1 taisho | Vitamin k1 tobishi;

- (KR) Korea, Republic of: Kaywan;
- (LB) Lebanon: Gnc Vitamin K | Konakion | Vitamin K;
- (LT) Lithuania: Kanavit | Konakion | Konakion mm | Konakion novum;
- (LV) Latvia: Kanavit;
- (MX) Mexico: Amidalak;
- (MY) Malaysia: Vitamin K | Vitamin K1;
- (NO) Norway: Konakion | Menadion | Vitacon | Vitacon campus;
- (NZ) New Zealand: Konakion mm | Vitamin K1;
- (PA) Panama: Fitomenadiona | Phytodon;
- (PE) Peru: Gnc Vitamin K | Vitamina k;
- (PH) Philippines: Hema-k;
- (PK) Pakistan: Vitamin K1;
- (PL) Poland: Kanavit;
- (PR) Puerto Rico: Mephyton | Phytonadione | Vitamin K;
- (PT) Portugal: Fitomenadiona | Kanakion | Kanakion MM | Kanakion Mm Pediatrico;
- (PY) Paraguay: Fitoquinona;
- (QA) Qatar: Kenadion | Konakion MM | Konakion MM Paediatric | Vitacon;
- (SA) Saudi Arabia: Amri k | Konakion;
- (SE) Sweden: Konakion;
- (SG) Singapore: Konakion mm;
- (SK) Slovakia: Kanavit;
- (TH) Thailand: K-vita | Konakion | Konakion mm;
- (TN) Tunisia: Vitamine k1;
- (TR) Turkey: Konakion | Selovita k1;
- (TW) Taiwan: Kaywan | Konakion mm | Phynadin k | Phytonadione | Vitamin K1;
- (UY) Uruguay: Kavit | Vitamina k;
- (VE) Venezuela, Bolivarian Republic of: Hemo k | Vitamina k | Vitamina K1;
- (ZA) South Africa: Konakion

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